

Module III: Optimization of Equality Constrained Continuous Functions

Optimization Techniques (24EST372)

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Outline

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Objectives

Understand the formulation of equality-constrained optimization problems.

Learn the Lagrange multiplier method for solving such problems.

Apply the method to theoretical and real-world optimization problems.

Practice through exercises and assignments, focusing on applications in AI and computer science.

General Form of Equality-Constrained Optimization

Optimize a continuous function subject to equality constraints:

Maximize/Minimize $f(\mathbf{x})$, $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$

Subject to $g_i(\mathbf{x}) = 0$, $i = 1, 2, \dots, m$

General Form of Equality-Constrained Optimization

Optimize a continuous function subject to equality constraints:

$$\text{Maximize/Minimize } f(\mathbf{x}), \quad \mathbf{x} = (x_1, x_2, \dots, x_n)^T$$

$$\text{Subject to } g_i(\mathbf{x}) = 0, \quad i = 1, 2, \dots, m$$

Goal: Find $\mathbf{x}^*$ that optimizes $f(\mathbf{x})$ while satisfying all constraints $g_i(\mathbf{x}) = 0$.

Lagrange Multiplier Method: Concept

Solves equality-constrained optimization by introducing Lagrange multipliers λ_i .

Forms the Lagrangian: $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x})$.

Finds stationary points by solving:

$$\nabla_{\mathbf{x}} \mathcal{L} = \mathbf{0}, \quad g_i(\mathbf{x}) = 0 \quad \forall i$$

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Finds stationary points by solving:

$$\nabla_{\mathbf{x}} \mathcal{L} = \mathbf{0}, \quad g_i(\mathbf{x}) = 0 \quad \forall i$$

For $f(\mathbf{x})$ and $g_i(\mathbf{x})$ differentiable, this yields critical points that may be optima.

Gradient Conditions

For a problem: Maximize/Minimize $f(\mathbf{x})$ subject to $g_i(\mathbf{x}) = 0$, the Lagrangian is:

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x})$$

Stationary points satisfy:

$$\begin{cases} \frac{\partial \mathcal{L}}{\partial x_j} = \frac{\partial f}{\partial x_j} + \sum_{i=1}^m \lambda_i \frac{\partial g_i}{\partial x_j} = 0, & j = 1, \dots, n \\ g_i(\mathbf{x}) = 0, & i = 1, \dots, m \end{cases}$$

Gradient Conditions

For a problem: Maximize/Minimize $f(\mathbf{x})$ subject to $g_i(\mathbf{x}) = 0$, the Lagrangian is:

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$$\begin{cases} \frac{\partial \mathcal{L}}{\partial x_j} = \frac{\partial f}{\partial x_j} + \sum_{i=1}^m \lambda_i \frac{\partial g_i}{\partial x_j} = 0, & j = 1, \dots, n \\ g_i(\mathbf{x}) = 0, & i = 1, \dots, m \end{cases}$$

This gives $n + m$ equations for n variables $\mathbf{x}$ and m multipliers $\boldsymbol{\lambda}$.

Step-by-Step Procedure for Lagrange Multipliers

To solve: Maximize/Minimize $f(\mathbf{x})$ subject to $g_i(\mathbf{x}) = 0$, $i = 1, \dots, m$:

Formulate Lagrangian: $\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x})$.

Compute Gradients: Calculate $\frac{\partial \mathcal{L}}{\partial x_j}$ and $\frac{\partial \mathcal{L}}{\partial \lambda_i} = g_i(\mathbf{x})$.

Set Gradients to Zero: Solve the system:

$$\begin{cases} \nabla_{\mathbf{x}} \mathcal{L} = \nabla f(\mathbf{x}) + \sum_{i=1}^m \lambda_i \nabla g_i(\mathbf{x}) = \mathbf{0} \\ g_i(\mathbf{x}) = 0, \quad i = 1, \dots, m \end{cases}$$

Solve System: Solve the $n + m$ equations for $\mathbf{x}$ and λ .

Classify Solutions: Use second-order conditions (Hessian of $\mathcal{L}$) or evaluate $f(\mathbf{x})$ to determine maxima, minima, or saddle points.

Verify Constraints: Ensure solutions satisfy $g_i(\mathbf{x}) = 0$.

Solved Problem 1: Standard Optimization

Maximize $f(x, y) = x^2 + 2y$ subject to:

$$g(x, y) = x^2 + y^2 - 4 = 0$$

Solved Problem 1: Standard Optimization

Maximize $f(x, y) = x^2 + 2y$ subject to:

$$g(x, y) = x^2 + y^2 - 4 = 0$$

Step 1: Form Lagrangian

$$\mathcal{L}(x, y, \lambda) = x^2 + 2y + \lambda(x^2 + y^2 - 4)$$

Step 2: Compute Gradients

$$\frac{\partial \mathcal{L}}{\partial x} = 2x + 2\lambda x = 0$$

$$\frac{\partial \mathcal{L}}{\partial y} = 2 + 2\lambda y = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = x^2 + y^2 - 4 = 0$$

Solved Problem 1: Solution

Step 3: Solve System From $\frac{\partial \mathcal{L}}{\partial x} = 2x(1 + \lambda) = 0$:

$$x = 0 \quad \text{or} \quad \lambda = -1$$

From $\frac{\partial \mathcal{L}}{\partial y} = 2 + 2\lambda y = 0$: $\lambda y = -1 \implies y = -\frac{1}{\lambda}$

Solved Problem 1: Solution

Step 3: Solve System From $\frac{\partial \mathcal{L}}{\partial x} = 2x(1 + \lambda) = 0$:

$$x = 0 \quad \text{or} \quad \lambda = -1$$

From $\frac{\partial \mathcal{L}}{\partial y} = 2 + 2\lambda y = 0$: $\lambda y = -1 \implies y = -\frac{1}{\lambda}$ **Case 1:** $x = 0$

$$y^2 - 4 = 0 \implies y = \pm 2$$

$$\lambda y = -1 \implies \lambda = -\frac{1}{y}$$

If $y = 2$, $\lambda = -\frac{1}{2}$, point: $(0, 2)$, $f(0, 2) = 4$

If $y = -2$, $\lambda = \frac{1}{2}$, point: $(0, -2)$, $f(0, -2) = -4$

Solved Problem 1: Solution

Step 3: Solve System From $\frac{\partial \mathcal{L}}{\partial x} = 2x(1 + \lambda) = 0$:

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If $y = 2$, $\lambda = -\frac{1}{2}$, point: $(0, 2)$, $f(0, 2) = 4$

If $y = -2$, $\lambda = \frac{1}{2}$, point: $(0, -2)$, $f(0, -2) = -4$

Case 2: $\lambda = -1$

$$y = -\frac{1}{-1} = 1, \quad x^2 + 1^2 = 4 \implies x^2 = 3 \implies x = \pm\sqrt{3}$$

Points: $(\sqrt{3}, 1)$, $f = 3 + 2 = 5$; $(-\sqrt{3}, 1)$, $f = 5$

Solved Problem 1: Solution

Step 3: Solve System From $\frac{\partial \mathcal{L}}{\partial x} = 2x(1 + \lambda) = 0$:

$$x = 0 \quad \text{or} \quad \lambda = -1$$

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$$y^2 - 4 = 0 \implies y = \pm 2$$

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If $y = 2$, $\lambda = -\frac{1}{2}$, point: $(0, 2)$, $f(0, 2) = 4$

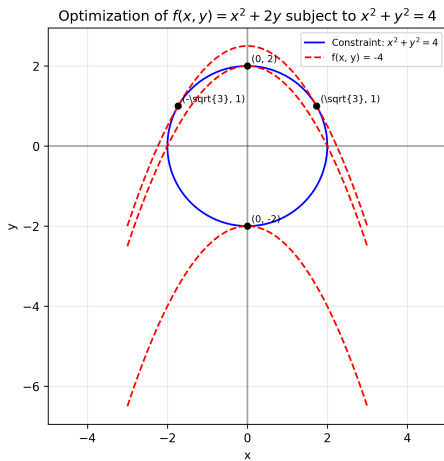
If $y = -2$, $\lambda = \frac{1}{2}$, point: $(0, -2)$, $f(0, -2) = -4$

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$$y = -\frac{1}{-1} = 1, \quad x^2 + 1^2 = 4 \implies x^2 = 3 \implies x = \pm\sqrt{3}$$

Points: $(\sqrt{3}, 1)$, $f = 3 + 2 = 5$; $(-\sqrt{3}, 1)$, $f = 5$ **Step 4: Classify** Evaluate f : Maxima at $(\sqrt{3}, 1)$, $(-\sqrt{3}, 1)$ ($f = 5$); minimum at $(0, -2)$ ($f = -4$).

figure



Solved Problem 2: Neural Network Weight Optimization

Minimize error $f(w_1, w_2) = w_1^2 + w_2^2$ (regularized loss) subject to:

$$g(w_1, w_2) = w_1 + 2w_2 - 3 = 0$$

Solved Problem 2: Neural Network Weight Optimization

Minimize error $f(w_1, w_2) = w_1^2 + w_2^2$ (regularized loss) subject to:

$$g(w_1, w_2) = w_1 + 2w_2 - 3 = 0$$

Step 1: Form Lagrangian

$$\mathcal{L}(w_1, w_2, \lambda) = w_1^2 + w_2^2 + \lambda(w_1 + 2w_2 - 3)$$

Step 2: Compute Gradients

$$\frac{\partial \mathcal{L}}{\partial w_1} = 2w_1 + \lambda = 0$$

$$\frac{\partial \mathcal{L}}{\partial w_2} = 2w_2 + 2\lambda = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = w_1 + 2w_2 - 3 = 0$$

Solved Problem 2: Solution

Step 3: Solve System From $\frac{\partial \mathcal{L}}{\partial w_1} = 2w_1 + \lambda = 0$: $\lambda = -2w_1$ From $\frac{\partial \mathcal{L}}{\partial w_2} = 2w_2 + 2\lambda = 0$:
 $w_2 + \lambda = 0 \implies \lambda = -w_2$ Equate: $-2w_1 = -w_2 \implies w_2 = 2w_1$ Constraint:
 $w_1 + 2(2w_1) = 3 \implies w_1 + 4w_1 = 3 \implies 5w_1 = 3 \implies w_1 = \frac{3}{5}$ Then: $w_2 = 2 \cdot \frac{3}{5} = \frac{6}{5}$

Solved Problem 2: Solution

Step 3: Solve System From $\frac{\partial \mathcal{L}}{\partial w_1} = 2w_1 + \lambda = 0$: $\lambda = -2w_1$ From $\frac{\partial \mathcal{L}}{\partial w_2} = 2w_2 + 2\lambda = 0$:
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Verify $\lambda = -2w_1 = -2 \cdot \frac{3}{5} = -\frac{6}{5}$ Check: $w_1 + 2w_2 = \frac{3}{5} + 2 \cdot \frac{6}{5} = \frac{3}{5} + \frac{12}{5} = 3$ Objective:
 $f = \left(\frac{3}{5}\right)^2 + \left(\frac{6}{5}\right)^2 = \frac{9}{25} + \frac{36}{25} = \frac{45}{25} = 1.8$

Solved Problem 2: Solution

Step 3: Solve System From $\frac{\partial \mathcal{L}}{\partial w_1} = 2w_1 + \lambda = 0$: $\lambda = -2w_1$ From $\frac{\partial \mathcal{L}}{\partial w_2} = 2w_2 + 2\lambda = 0$:
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Verify $\lambda = -2w_1 = -2 \cdot \frac{3}{5} = -\frac{6}{5}$ Check: $w_1 + 2w_2 = \frac{3}{5} + 2 \cdot \frac{6}{5} = \frac{3}{5} + \frac{12}{5} = 3$ Objective:
 $f = \left(\frac{3}{5}\right)^2 + \left(\frac{6}{5}\right)^2 = \frac{9}{25} + \frac{36}{25} = \frac{45}{25} = 1.8$ **Step 5: Classify** Hessian of f : $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, positive definite,
confirms minimum. **Solution:** $(w_1, w_2) = \left(\frac{3}{5}, \frac{6}{5}\right)$, $f = 1.8$. **Interpretation:** Optimal weights
minimize error while satisfying the constraint (e.g., linear activation balance in a neural network).

Solved Problem 3: Standard Optimization

Maximize $f(x, y) = xy$ subject to:

$$g(x, y) = x^2 + y^2 - 1 = 0$$

Solved Problem 3: Standard Optimization

Maximize $f(x, y) = xy$ subject to:

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Step 1: Form Lagrangian

$$\mathcal{L}(x, y, \lambda) = xy + \lambda(x^2 + y^2 - 1)$$

Step 2: Compute Gradients

$$\frac{\partial \mathcal{L}}{\partial x} = y + 2\lambda x = 0$$

$$\frac{\partial \mathcal{L}}{\partial y} = x + 2\lambda y = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = x^2 + y^2 - 1 = 0$$

Solved Problem 3: Solution

Step 3: Solve System From $\frac{\partial \mathcal{L}}{\partial x} = y + 2\lambda x = 0$: $y = -2\lambda x$ From $\frac{\partial \mathcal{L}}{\partial y} = x + 2\lambda y = 0$:
 $x = -2\lambda y$

Solved Problem 3: Solution

Step 3: Solve System From $\frac{\partial \mathcal{L}}{\partial x} = y + 2\lambda x = 0$: $y = -2\lambda x$ From $\frac{\partial \mathcal{L}}{\partial y} = x + 2\lambda y = 0$:
 $x = -2\lambda y$ **Substitute and Solve** If $x \neq 0$, $y = -2\lambda x$ in $x = -2\lambda y$:

$$x = -2\lambda(-2\lambda x) = 4\lambda^2 x$$

$$x(4\lambda^2 - 1) = 0 \implies \lambda = \pm \frac{1}{2} \quad (\text{since } x \neq 0)$$

Solved Problem 3: Solution

Step 3: Solve System From $\frac{\partial \mathcal{L}}{\partial x} = y + 2\lambda x = 0$: $y = -2\lambda x$ From $\frac{\partial \mathcal{L}}{\partial y} = x + 2\lambda y = 0$:
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$$x = -2\lambda(-2\lambda x) = 4\lambda^2 x$$

$$x(4\lambda^2 - 1) = 0 \implies \lambda = \pm \frac{1}{2} \quad (\text{since } x \neq 0)$$

Case 1: $\lambda = \frac{1}{2}$

$$y = -2 \cdot \frac{1}{2} x = -x, \quad x^2 + (-x)^2 = 1 \implies x^2 + x^2 = 1 \implies 2x^2 = 1 \implies x = \pm \frac{\sqrt{2}}{2}$$

Points: $\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$, $f = \frac{\sqrt{2}}{2} \cdot \left(-\frac{\sqrt{2}}{2}\right) = -\frac{1}{2}$; $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$, $f = -\frac{1}{2}$

Solved Problem 3: Solution

Step 3: Solve System From $\frac{\partial \mathcal{L}}{\partial x} = y + 2\lambda x = 0$: $y = -2\lambda x$ From $\frac{\partial \mathcal{L}}{\partial y} = x + 2\lambda y = 0$:
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Points: $\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$, $f = \frac{\sqrt{2}}{2} \cdot \left(-\frac{\sqrt{2}}{2}\right) = -\frac{1}{2}$; $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$, $f = -\frac{1}{2}$ **Case 2:** $\lambda = -\frac{1}{2}$

$$y = -2 \cdot \left(-\frac{1}{2}\right) x = x, \quad x^2 + x^2 = 1 \implies 2x^2 = 1 \implies x = \pm \frac{\sqrt{2}}{2}$$

Points: $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$, $f = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{2} = \frac{1}{2}$; $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$, $f = \frac{1}{2}$

Solved Problem 3: Solution

Step 3: Solve System From $\frac{\partial \mathcal{L}}{\partial x} = y + 2\lambda x = 0$: $y = -2\lambda x$ From $\frac{\partial \mathcal{L}}{\partial y} = x + 2\lambda y = 0$:
 $x = -2\lambda y$ **Substitute and Solve** If $x \neq 0$, $y = -2\lambda x$ in $x = -2\lambda y$:

$$x = -2\lambda(-2\lambda x) = 4\lambda^2 x$$

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Case 1: $\lambda = \frac{1}{2}$

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Points: $\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$, $f = \frac{\sqrt{2}}{2} \cdot \left(-\frac{\sqrt{2}}{2}\right) = -\frac{1}{2}$; $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$, $f = -\frac{1}{2}$ **Case 2:** $\lambda = -\frac{1}{2}$

$$y = -2 \cdot \left(-\frac{1}{2}\right) x = x, \quad x^2 + x^2 = 1 \implies 2x^2 = 1 \implies x = \pm \frac{\sqrt{2}}{2}$$

Points: $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$, $f = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{2} = \frac{1}{2}$; $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$, $f = \frac{1}{2}$ **Step 4: Classify** Evaluate f :

Maxima at $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$, $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$ ($f = \frac{1}{2}$); minima at $\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$, $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$ ($f = -\frac{1}{2}$).

Solved Problem 4: Neural Network Hyperparameter Tuning

Minimize $f(w_1, w_2) = w_1^2 + w_2^2 + w_1 w_2$ (regularized loss) subject to:

$$g(w_1, w_2) = 2w_1 + w_2 - 4 = 0$$

Solved Problem 4: Neural Network Hyperparameter Tuning

Minimize $f(w_1, w_2) = w_1^2 + w_2^2 + w_1 w_2$ (regularized loss) subject to:

$$g(w_1, w_2) = 2w_1 + w_2 - 4 = 0$$

Step 1: Form Lagrangian

$$\mathcal{L}(w_1, w_2, \lambda) = w_1^2 + w_2^2 + w_1 w_2 + \lambda(2w_1 + w_2 - 4)$$

Step 2: Compute Gradients

$$\frac{\partial \mathcal{L}}{\partial w_1} = 2w_1 + w_2 + 2\lambda = 0$$

$$\frac{\partial \mathcal{L}}{\partial w_2} = 2w_2 + w_1 + \lambda = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 2w_1 + w_2 - 4 = 0$$

Solved Problem 4: Solution

Step 3: Solve System From $\frac{\partial \mathcal{L}}{\partial w_1} = 2w_1 + w_2 + 2\lambda = 0$: $w_2 = -2w_1 - 2\lambda$ From $\frac{\partial \mathcal{L}}{\partial w_2} = 2w_2 + w_1 + \lambda = 0$: $2w_2 + w_1 + \lambda = 0$ Constraint: $2w_1 + w_2 = 4$

Solved Problem 4: Solution

Step 3: Solve System From $\frac{\partial \mathcal{L}}{\partial w_1} = 2w_1 + w_2 + 2\lambda = 0$: $w_2 = -2w_1 - 2\lambda$ From $\frac{\partial \mathcal{L}}{\partial w_2} = 2w_2 + w_1 + \lambda = 0$: $2w_2 + w_1 + \lambda = 0$ Constraint: $2w_1 + w_2 = 4$ Substitute $w_2 = 4 - 2w_1$ into the first equation:

$$2w_1 + (4 - 2w_1) + 2\lambda = 0 \implies 4 + 2\lambda = 0 \implies \lambda = -2$$

Substitute $\lambda = -2$ into the second equation:

$$2w_2 + w_1 - 2 = 0$$

With $w_2 = 4 - 2w_1$:

$$2(4 - 2w_1) + w_1 - 2 = 0 \implies 8 - 4w_1 + w_1 - 2 = 0 \implies -3w_1 + 6 = 0 \implies w_1 = 2$$

Then: $w_2 = 4 - 2 \cdot 2 = 0$

Solved Problem 4: Solution

Step 3: Solve System From $\frac{\partial \mathcal{L}}{\partial w_1} = 2w_1 + w_2 + 2\lambda = 0$: $w_2 = -2w_1 - 2\lambda$ From $\frac{\partial \mathcal{L}}{\partial w_2} = 2w_2 + w_1 + \lambda = 0$: $2w_2 + w_1 + \lambda = 0$ Constraint: $2w_1 + w_2 = 4$ Substitute $w_2 = 4 - 2w_1$ into the first equation:

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Then: $w_2 = 4 - 2 \cdot 2 = 0$ **Step 4: Verify** Check constraint: $2 \cdot 2 + 0 = 4$ Objective:

$$f = 2^2 + 0^2 + 2 \cdot 0 = 4$$

Solved Problem 4: Solution

Step 3: Solve System From $\frac{\partial \mathcal{L}}{\partial w_1} = 2w_1 + w_2 + 2\lambda = 0$: $w_2 = -2w_1 - 2\lambda$ From $\frac{\partial \mathcal{L}}{\partial w_2} = 2w_2 + w_1 + \lambda = 0$: $2w_2 + w_1 + \lambda = 0$ Constraint: $2w_1 + w_2 = 4$ Substitute $w_2 = 4 - 2w_1$ into the first equation:

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Substitute $\lambda = -2$ into the second equation:

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With $w_2 = 4 - 2w_1$:

$$2(4 - 2w_1) + w_1 - 2 = 0 \implies 8 - 4w_1 + w_1 - 2 = 0 \implies -3w_1 + 6 = 0 \implies w_1 = 2$$

Then: $w_2 = 4 - 2 \cdot 2 = 0$ **Step 4: Verify** Check constraint: $2 \cdot 2 + 0 = 4$ Objective:

$f = 2^2 + 0^2 + 2 \cdot 0 = 4$ **Step 5: Classify** Hessian of f : $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $\det = 2 \cdot 2 - 1 \cdot 1 = 3 > 0$, and

$\partial^2 f / \partial w_1^2 = 2 > 0$, so positive definite (minimum). **Solution:** $(w_1, w_2) = (2, 0)$, $f = 4$.

Interpretation: Optimal hyperparameters (e.g., learning rate $w_1 = 2$, regularization $w_2 = 0$) minimize loss within the computational budget.

Application: Resource Allocation

Maximize production $f(x, y) = 30x + 50y - 2x^2 - 2y$ subject to:

$$g(x, y) = x + y - 10 = 0$$

Formulate the Lagrangian and solve using Lagrange multipliers.

Interpret the solution in the context of resource allocation (e.g., labor and capital).

Analyze the impact of changing the constraint constant (e.g., to 11).

Exercise 1

Maximize $f(x, y) = xy$ subject to:

$$x^2 + y^2 = 8$$

Use Lagrange multipliers to find the critical points.

Classify the points as maxima or minima.

Verify the solution satisfies the constraint.

Exercise 2

Minimize $f(x, y) = x^2 + y^2$ subject to:

$$x + y = 4$$

Solve using Lagrange multipliers.

Confirm the solution is a minimum using the Hessian.

Interpret the result in a geometric context.

Assignment 1 (Individual)

Maximize $f(x, y) = 4x + 3y$ subject to:

$$x^2 + y^2 = 25$$

Use Lagrange multipliers to solve.

Submit a report including:

- Detailed solution steps.

- Classification of critical points.

- Graphical representation of the constraint and objective.

Discuss the solution's relevance to resource optimization.

Assignment 2 (Team Work)

Optimize neural network loss $f(w_1, w_2) = w_1^2 + 2w_2^2$ subject to:

$$w_1 + w_2 = 5$$

Form a team of 3–4 students.

Solve using Lagrange multipliers.

Prepare a presentation including:

- Solution process and verification.

- Interpretation in the context of neural network training.

- Sensitivity of the solution to constraint changes.

Deadline: One week from today.

References

H. A. Taha, *Operations Research: An Introduction*, 10th ed., Pearson, 2017.

Kanti Swarup, P.K. Gupta, Man Mohan, *Operations Research*, 20th ed., Sultan Chand and Sons, 2022.

S. S. Rao, *Engineering Optimization: Theory and Practice*, 4th ed., Wiley, 2009.

M. S. Bazaraa, H. J. Sherali, C. M. Shetty, *Nonlinear Programming: Theory and Algorithms*, 3rd ed., Wiley, 2013.

NPTEL: Prof. S. S. Rao, https://onlinecourses.nptel.ac.in/noc20_me46/preview

Thank You

Questions?